# Soham Karpe, M. Eng.

*AI Systems Engineer – Computer Vision / Machine Vision*

+49 15510686539 karpesoham@gmail.com

Steinweg 17, 93413 Cham, Germany | Willing to relocate

Date of Birth: 03.02.1998, India Driving License: Class B Portfolio

## PROFILE

AI-focused engineer specializing in **computer vision, machine learning, and industrial automation systems**. Experienced in developing **AI-based visual inspection solutions, real-time image processing pipelines, and deep learning models using C++, Python, OpenCV, PyTorch, TensorFlow, and YOLO**, as well as classical machine learning techniques using **scikit-learn**. Skilled in integrating **camera systems, embedded hardware, and PyQt-based HMI interfaces** for industrial environments. Passionate about applying AI and computer vision to solve real-world engineering challenges in **automation and intelligent production systems**.

## PROFESSIONAL EXPERIENCE

**AI Systems Engineer (Intern) – Computer Vision / Machine Vision** Jul 2024 – Present | Roding, Germany

• Developed AI-based computer vision pipelines for industrial visual inspection using C++, Python, OpenCV, PyTorch, and YOLO.
• Built end-to-end machine learning workflows, including dataset creation, model training, evaluation, and real-time inference deployment.
• Designed and optimized real-time image processing systems for defect detection and automated visual analysis.
• Collaborated with software, hardware, and automation teams to integrate AI models with industrial systems and production environments.

**Junior Software Developer (Werkstudent)** Mar 2024 – Jun 2024 | Roding, Germany
**mech-tron GmbH & Co. KG**

• Developed anomaly detection pipelines using PyTorch and TensorFlow.
• Built a facial recognition system integrating computer vision models with Python backend services and C++.
• Improved ML workflow reliability through data validation, automation, and edge optimization.

**Software Engineer** Jul 2020 – Jul 2022 | Pune, India
**Evoka Technologies Pvt. Ltd.**

• Contributed to software development and early product prototyping during the company's pre-incorporation phase.
• Worked as a Software Engineer post-incorporation on internal projects, contributing to development, testing, and module improvements.
• Gained hands-on experience in a startup environment, building product thinking skills and working with agile practices.

**Internship & Bachelor Thesis** Sep 2019 – Jun 2020 | Pune, India
**Profiroll Technologies India Pvt. Ltd.**

• Designed and optimized a compact hydraulic straightening machine to improve mechanical alignment and operational efficiency.
• Performed system modeling and engineering analysis to evaluate mechanical performance and structural stability.

## EDUCATION

**M.Eng. Artificial Intelligence for Smart Sensors and Actuators** Oct 2022 – Mar 2026 | Cham, Germany
**Technische Hochschule Deggendorf (Deggendorf Institute of Technology- DIT)**

Artificial Intelligence, Model Based Function Engineering, Machine Learning and Deep Learning, Robotics and Autonomous Systems, Big Data, Computer Vision, System Design and Intercommunication, Embedded Systems.

**Master Thesis:** mech-tron GmbH & Co. KG, Roding
**Design and Development of a Real-Time AI-Based Visual Inspection System (Best Thesis Nominated)**
AI Visual Inspection • Computer Vision Pipelines • Camera Hardware Integration • Qt based HMI Development • Industrial Automation • Image Processing • Defect Detection • Camera Calibration • Model Training & HMI Configuration • Real-Time System Setup

**Bachelor of Engineering** Jul 2016 – Jun 2020 | Pune, India
Pune University, India
**Average Grade: 2.3**
Programming Fundamentals, Engineering Mathematics, Data Structures (Fundamentals), Control Systems, Mechatronics Systems, FEA, Applied Mechanics, Strength of Materials, Dynamics of Machinery

## PROJECTS

**Liveness Detection using Computer Vision (Image Processing)** Apr 2024 – Jul 2024 | DIT, HiWi
- Developed and evaluated a YOLOv8-based computer vision model for human liveness detection, designing the custom dataset, training pipeline, and experimental setup. Achieved 91% accuracy through systematic model tuning, data augmentation, and performance benchmarking across varied real-world conditions.

**Building Height Detection using Advanced Intelligence System** Mar 2023 – Jul 2023 | DIT, HiWi
- Developed MATLAB and Python machine learning models for building height estimation, designing feature extraction, visualization, and real-time processing workflows to enable accurate analysis for architectural and urban scenarios.

**Pick-and-Place Scara Robot using Gazebo Simulation** Mar 2024 – Jul 2024 | DIT, Case Study
- Designed and simulated a pick-and-place SCARA robot in Gazebo using ROS, implementing kinematics, motion planning, and control algorithms to enable precise object manipulation for industrial automation scenarios.

## KNOWLEDGE AND TECHNICAL SKILLS

**Programming Languages**
C++, C, Python, Java

**Embedded & GUI Development**
Embedded Programming, Qt Framework

**Data Tools**
**Tools & Platforms:** Docker, Git/GitHub, CI/CD, Python Automation Scripts
**Cloud:** Google Cloud (BigQuery), AWS (Basics)

**Languages**
German- B1 (actively improving, goal C1)
English - B2
Hindi & Marathi- Native

**Prompt Engineering**
LLM Integration, Prompt Engineering, AI-assisted Workflows

**Embedded / Edge Deployment:**
Raspberry Pi, ESP32, GPIO, MQTT, Edge Inference, Arduino Mega, daheng imaging camera, Basler Camera

**Productivity Tools & Visualization**
Visual Studio Code, Git/GitHub, Linux, PowerShell, QEMU, Matplotlib, Seaborn

**AI Skills**
YOLOv8, PyTorch, TensorFlow, scikit-learn, Computer Vision, Dataset Preparation, Model Optimization, Real-Time Inference, Hyperparameter Tuning

**Soft Skills**
Analytical Thinking, Problem Solving, Communication, Team Collaboration, Initiative

## CERTIFICATION

**Machine Learning with Python** • **Introduction to Deep Learning & Neural Networks with Keras** • **AWS Certified Machine Learning - Specialty** • **AWS AI Practitioner Certification Prep** • **Goethe-Institut A2 and B1 (German Super Intensive Course)** • **: Pursuing -** | **Machine Learning Specialization (Andrew Ng)** | **Computer Vision with TensorFlow** | **Advanced Machine Learning on Google Cloud Specialization** | **MathWorks Computer Vision Engineer Professional Certificate**

## INTERESTS

Strong interest in developing machine learning and AI systems for data analysis, model evaluation, and performance optimization. Experienced in deep learning techniques for text, image and video understanding and intelligent decision-making. Passionate about building scalable ML pipelines and deploying data-driven solutions to real-world problems.

Cham, 12.03.2026